NNN-P-385C
April 15, 1978
SUPERSEDING
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FEDERAL SPECIFICATION

PIPET, BLOOD CHEMISTRY (MICRO-FOLIN)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a graduated serological pipet suitable for use in micro techniques for estimation of blood sugar by the Folin method.

1.2 Classification.

1.2.1 Capacity. Pipets covered by this specification shall be of the following capacities, as specified (see 6.1).

Capacities:

0.1 milliliter (ml).
0.2 ml.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6640

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

C 148 - Polariscopic Examination of Glass Containers.
C 162 - Definition of Terms Relating to Glass and Glass Containers.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Bureau of Standards Internal Report:

NBSIR 74-461 - The Calibration of Small Volumetric Laboratory Glassware.

(Application for copies should be addressed to the National Technical Information Service (NTIS), Springfield, VA 22151.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. Pipet shall be commercially available borosilicate glass.

3.2 Annealing. When examined with a polariscope as specified in 4.4.1, there shall be no ring strain greater than temper No. 1 except as permitted for the tempered tip in 3.6.

3.3 Design. The pipet shall be a microblood pipet of Folin design. The cross-section of the pipet shall be circular except that the mouth end shall have an expanded bore with a wall thickness of not less than 1 millimeter (mm) to hold a cotton plug. The pipet shall conform to ASTM E 193 (figure 2) as to dimensions and tolerances (see table I).

TABLE I. Requirements for serological pipets

Capacity, ml.	Tolerance, ml.	Color coding band
0.1	± 0.001	Blue
0.2	± 0.002	Orange

3.3.1 Tip. The end of the tip shall be perpendicular to the longitudinal axis of the pipet. The outside edge of the top shall be beveled slightly and the bevel shall be ground or fire-polished.

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3.4 Capacity and graduations. The pipet shall be graduated to contain 0.1 ml and 0.2 ml. Markings shall fall within the tolerances as indicated in figure 1. The graduation markings of the pipet shall be tested for capacity accuracy in accordance with 4.4.2.

3.4.1 Markings. All markings shall be permanent and legible. Graduation markings shall be applied by one of the following methods---etched and filled with a permanent pigment; engraved and filled with a permanent pigment; by application of a stain fired into the glass without etching; or by application of enamel fired onto the glass without etching. There shall be no change in the appearance of the markings when tested for permanency as specified in 4.4.3.

3.4.2 Identification. The pipet shall be marked with the manufacturer's name or trademark, its capacity, the symbol "TC", and the temperature of calibration (20°C).

3.5 Color coding. The pipet shall be marked (see table I) with one or two color bands as indicated. The color of the band shall be bright and distinct. When tested as specified in 4.4.3, the color shall show no damage other than loss of gloss. Banding shall meet the alkali and acid tests as specified in 4.4.3.1 and 4.4.3.2, respectively.

3.6 Tempered tip. The pipet shall have a tempered tip. When tested in index oil which matches the refractive index of the glass being used and in accordance with 4.4.4, the temper shall be between 75 and 220 millimicrons.

3.7 Workmanship. The pipet shall be free from defects which detract from its appearance or impair serviceability. The pipet shall be transparent, clean, dry, and free of foreign material. The bore of the pipet shall be free of constrictions and shall permit complete emptying of fluids (see 4.3.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table II.

TABLE II. Sampling for examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	2.5
Minor defects	II	4.0
For dimensional examination	S-3	2.5

4.2.2 For tests. Sampling for tests shall be in accordance with MIL-STD-105. Unit of product shall be pipet.

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TABLE III. Testing requirements

Characteristics	Requirement paragraph	Test procedure	Inspection level	AQL (percent defective)
Annealing	3.2	4.4.1	S-2	1.0
Capacity	3.4	4.4.2	S-2	1.0
Permanency of marking		4.4.3		
Alkali test	3.5	4.4.3.1	S-3	4.0
Acid test	3.5	4.4.3.2	S-3	4.0
Tempered tips	3.6	4.4.4	S-2	1.0

4.3 Examination. The pipets shall be examined to determine compliance with all the requirements contained in the specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with table IV.

TABLE IV. Classification for defects

Categories	Defects 1/
Major	
101	Pipet not circular or not free of distortion or obstructions.
102	Pipet not free of burrs, protrusions, or sharp edges.
103	Pipet not free of cracks or chippings.
104	Pipet not free of stones, seeds, striae, or blisters.
105	Pipet glass not clear, or not free of cloudiness.
106	Pipet not free of grease, dirt, or foreign matter.
107	Capacity mark on pipet not present, not perpendicular to longitudinal axis, or not distinct.
108	Color band on pipet incorrect, or not present.
109	Mouthpiece and top of pipet not beveled, grounded or fire-polished.
110	Marking on pipet not complete, correct, distinct or legible including manufacturer's identification.
Minor	
201	Color band of pipet not complete.
202	Pipet glass not free of scratch marks.
203	Pipet not free of bubbles.

1/ Examination shall not be restricted to the classified defects listed above. Definition of glass defects shall be in accordance with ASTM C 162.

4.3.2 Dimensional examination. The pipet shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Distilled water and reagent grade chemicals shall be used throughout the tests. Tests shall be conducted as follows:

4.4.1 Annealing. Pipets shall be tested in accordance with method B of ASTM C 148.

4.4.2 Capacity. The capacity of the pipets shall be determined in accordance with the procedures of NBSIR 74-461.

4.4.3 Permanency of marking. A different pipet shall be used for each of the following tests.

4.4.3.1 Resistance to alkali. The pipet shall be completely immersed in 1N sodium hydroxide solution at room temperature, covered, boiled for 1/2 hour and allowed to stand in the alkali for an additional hour. The pipet shall then be removed, thoroughly rinsed with distilled water, and dried by rubbing vigorously with a cloth.

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4.4.3.2 Resistance to acid. The pipet shall be completely immersed in concentrated sulfuric acid (specific gravity 1.84) at room temperature and allowed to stand 1 hour. The pipet shall then be removed, thoroughly rinsed with distilled water, and dried by rubbing vigorously with a cloth.

4.4.4 Tempered tip. Tip shall be tested in accordance with method B of ASTM C 148. The tips shall be immersed in a solution whose index of refraction matches that of the glass.

4.5 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements in section 5. Defects shall be scored in accordance with table V. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of containers offered for examination at one time. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

TABLE V. Preparation for delivery defects

Examine	Defects
Material	Any component missing, damaged, or not as specified.
Container	Not as specified.
Workmanship	Inadequate application of components, container bulged or distorted.
Content	Not as specified.
Marking	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, or Commercial as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unit container. Six (6) pipets of the same capacity shall be packaged in a shrink pack tray of the type normally used for the item.

5.1.1.2 Intermediate container. Two (2) trays of pipets of the same capacity shall be intermediately packaged in a fiberboard box. The box shall be closed with adhesive, staples, or tape to secure the flaps.

5.1.2 Commercial. The pipets shall be preserved and packaged in accordance with normal commercial practice. The complete package shall be designed to protect the item against damage during shipment, handling and storage.

5.2 Packing. Packing shall be level A, or Commercial as specified (see 6.1).

5.2.1 Level A. Twenty-four (24) intermediate containers of pipets of the same capacity shall be packed in a weather resistant, corrugated fiberboard box. The box shall be closed with adhesive, staples or tape to secure the flaps. The box shall be sealed to enhance water resistance. The box shall be banded with metal or plastic strapping or tape for reinforcement.

5.2.2 Commercial. The packaged pipets shall be packed in fiberboard boxes to insure safe delivery at destination, to provide for safe redistribution by the initial receiving activity and shall be acceptable by common carrier under National Motor Freight Classification or Uniform Freight Classification.

5.3 Unitization. When shipments to Government depots are full car or truckload, the shipping containers shall be unitized for shipment and handling in accordance with normal commercial practice. The unitized load shall not exceed 2,500 pounds in weight, 63 inches in height, 56 inches in length, and 45 inches in width.

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5.4 Marking. The unit, intermediate and shipping containers and unitized loads, as applicable, shall be marked as specified in the contract or order (see 6.1).

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Capacity required (see 1.2.1).
(c) Packaging and packing required (see 5.1 and 5.2).
(d) Marking required (see 5.4).

MILITARY CUSTODIANS:

Army - MU
Navy - MS
Air Force - 03
DSA - DM

Review activity:

DSA - DM

Preparing activity:

GSA-FSS

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS
HEW, FDA, NIH
VA-DMS
AGR-ARS

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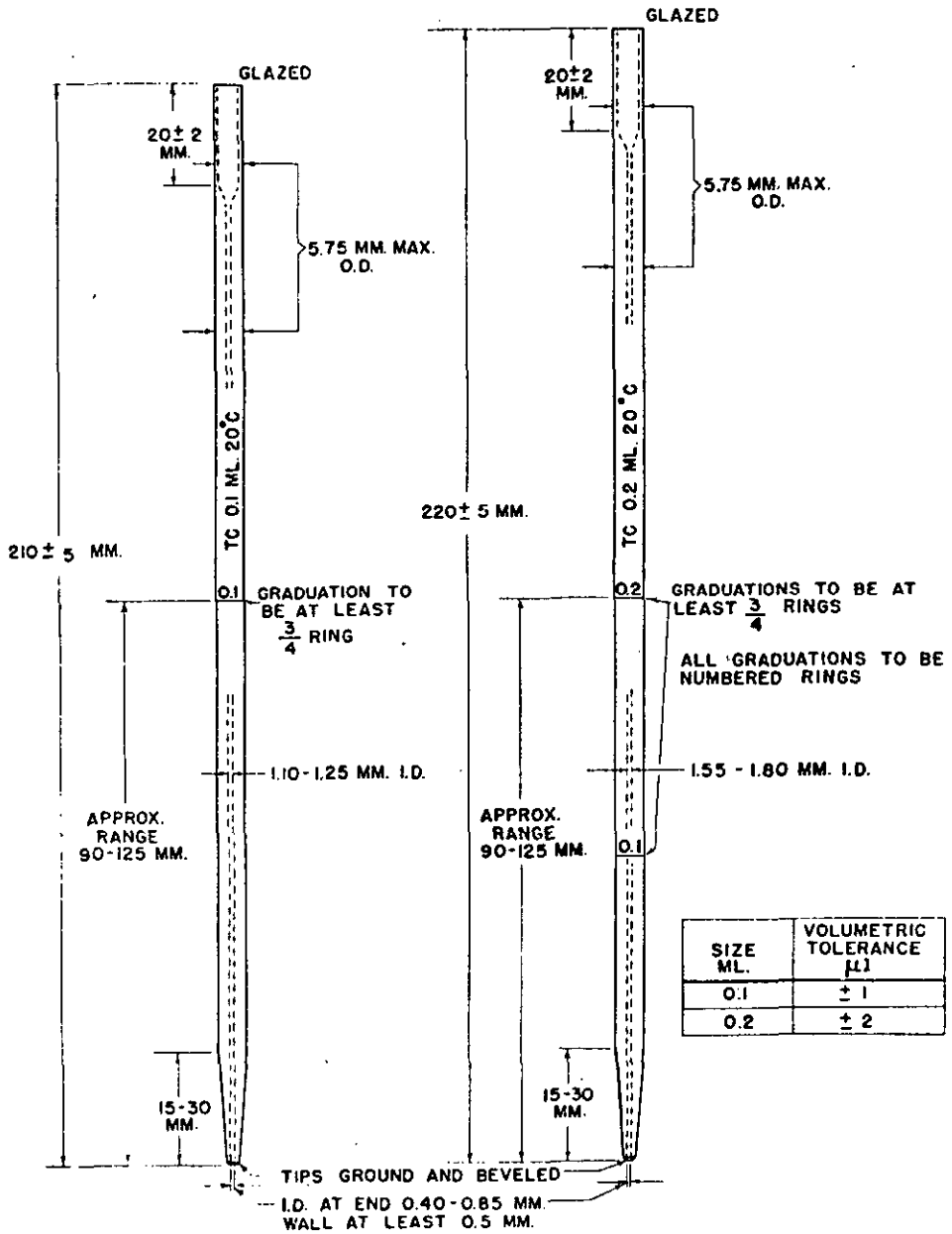
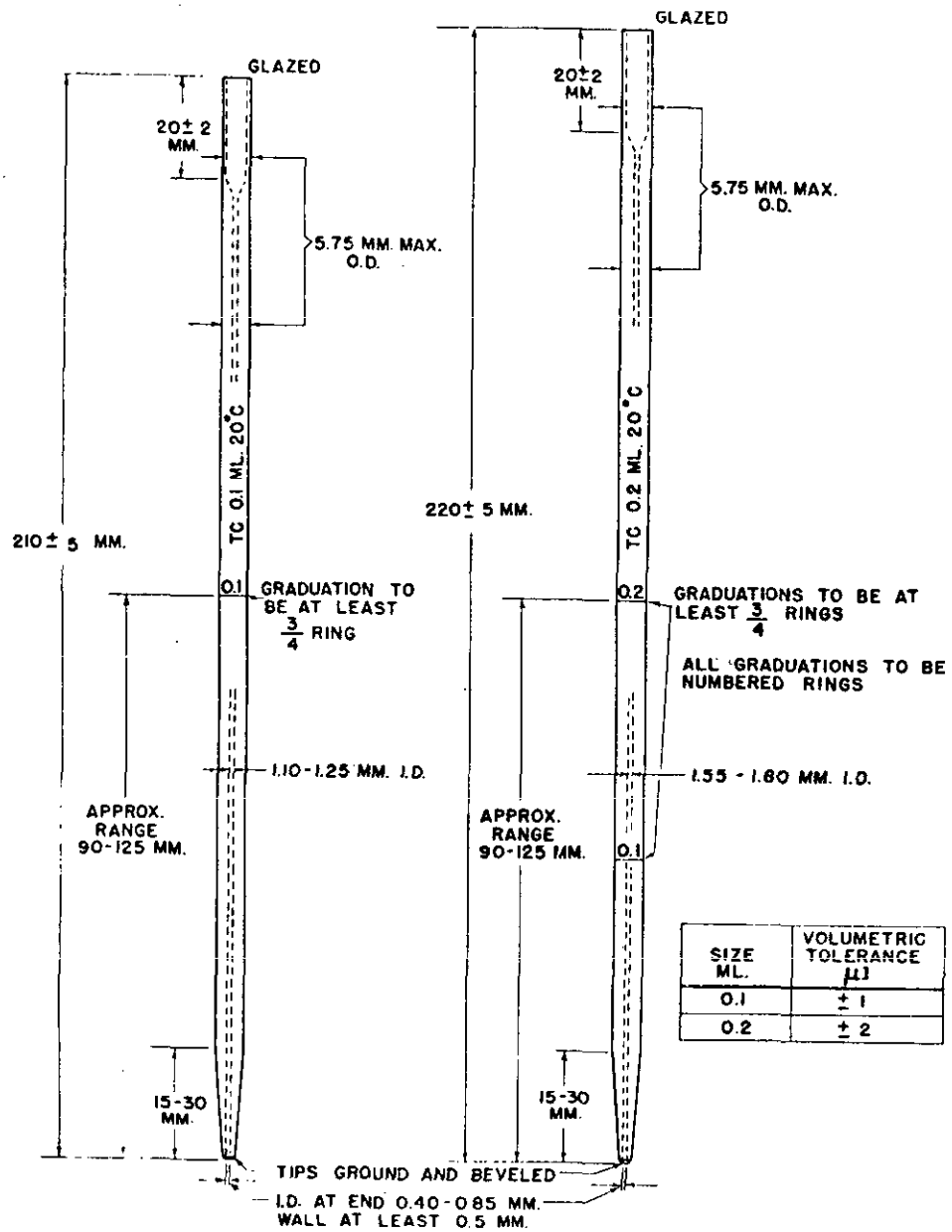


FIGURE 1. Micropipets, Folin-Type.

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SIZE ML.	VOLUMETRIC TOLERANCE (µl)
0.1	± 1
0.2	± 2

FIGURE 2 - Micropipets, Folin-Type

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 60 cents each.

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